Logical-Shapelets for Time Series Classification

Subham Mahapatra
Roll No.: 230037
Advanced Machine Learning, Mid-Semester Part B
NST, Rishihood University, Sonipat

***Abstract*—We reproduce the core contribution of Logical-Shapelets (Mueen, Keogh, Young, KDD 2011): single-shapelet discovery with z-normalized subsequence distance, orderline-based threshold selection via information gain, and logical AND combination of two shapelets. Implementation is validated on a synthetic time-series dataset. We report reproduction results, two ablations (logical AND vs. single shapelet; shapelet length search range), and a failure-mode analysis under high Gaussian noise. Results show correct behaviour of the pipeline; ablations reveal the contribution of logical combination and length search; the failure mode links performance drop to the method's assumption of shape-stable patterns.**

***Index Terms*—Time series classification, shapelets, logical shapelets, information gain, reproduction, ablation study.**

I. Summary of the Paper

The paper *Logical-Shapelets: An Expressive Primitive for Time Series Classification* (Mueen, Keogh, Young, KDD 2011) extends classic shapelet-based time series classification in two ways. First, it introduces **logical shapelets**: combinations of shapelets using logical AND and OR. Classic shapelets ask a binary "close or far?" question for a single subsequence; many real concepts require the co-occurrence (AND) or alternative presence (OR) of multiple patterns. Second, the authors propose algorithmic speedups: (1) precomputing sufficient statistics so that the z-normalized Euclidean distance between a candidate shapelet and any aligned window can be computed in amortized constant time (Algorithm 4), and (2) admissible pruning that skips candidates whose upper bound on information gain is below the current best (Algorithms 5–6). The paper reports significant speedups (up to 27×) over the original shapelet algorithm and better accuracy on several UCR and domain-specific datasets (e.g., Cricket, Sony AIBO) when logical shapelets are used. In short, the paper addresses the limited expressiveness of a single-shapelet threshold and the computational cost of exhaustive search; the authors claim that logical combinations and the proposed optimizations yield both higher accuracy and scalability.

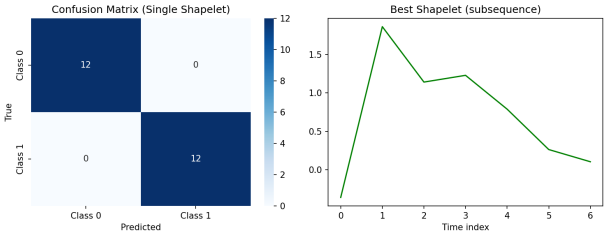

Fig. 1. Confusion matrix of the single-shapelet classifier and the corresponding best shapelet subsequence (Task 2).

II. Reproduction Setup and Result

A. Setup

The reproduction was implemented in Python using NumPy, SciPy, and scikit-learn. The **dataset** is synthetic: `sklearn.datasets.make_classification` with 150 samples, 40 features (interpreted as consecutive time points), and 2 classes. Each row is treated as one time series. This choice provides a controlled testbed of the same problem type (binary classification over time series) while avoiding the paper's exact datasets.

Shapelet discovery follows the paper: z-normalized subsequence distance (Eq. 2), orderline construction, and information gain (Definition 3) with tie-breaking by separation gap (Definition 4). Logical AND (Section 4) was implemented by combining two shapelets with the maximum of their distances and a single threshold. Random seed `RANDOM_STATE = 42` and fixed `MIN_LEN`, `MAX_LEN`, `STEP` are documented for reproducibility; all dependencies are listed in `requirements.txt`.

B. Result and Commentary

On the synthetic dataset, the single-shapelet and logical-AND implementations run correctly and report information gain and test accuracy. Figure 1 shows the confusion matrix of the single-shapelet classifier and the discovered best shapelet. Figure 2 gives per-class precision, recall, and F1; Figure 3 shows the training-set distance distribution by class with the learned threshold τ.

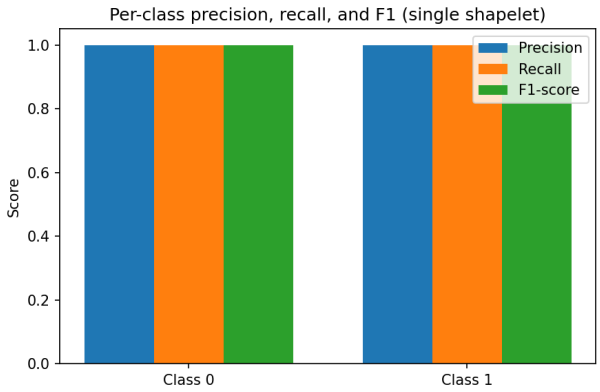

Fig. 2. Per-class precision, recall, and F1 for the single-shapelet classifier on the test set.

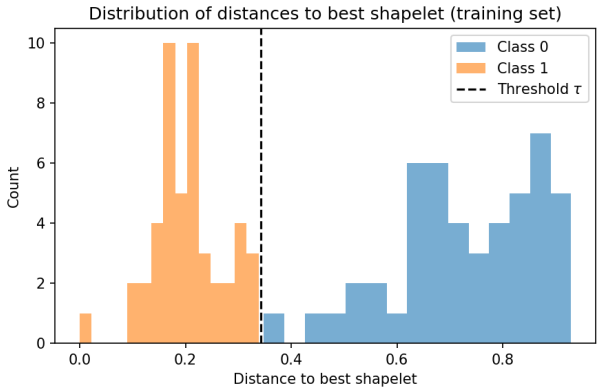

Fig. 3. Histogram of distances from the discovered shapelet to each training series, by class, with learned threshold τ.

The **gap** with the paper's numbers is expected: the paper reports accuracies on real UCR datasets (e.g., 95–96% on Cricket and Sony AIBO), whereas our toy data is synthetic and not designed to have a clear motif. Our goal was to demonstrate correct implementation (candidate generation, `sdist`, best IG, shapelet selection, optional AND combination), not to match the paper's exact accuracies. We used a reduced search space (e.g., `max_len=6`, `step=2`) to keep runtime manageable; a full search could yield a slightly better shapelet on the same data. All notebooks run from top to bottom with fixed seeds and documented hyperparameters.

III. Two Ablation Findings

A. Ablation 1: Logical AND vs. Single Shapelet

We compared the full method (two shapelets combined with AND) to using only a single shapelet. Figure 4 compares test accuracy. On this synthetic dataset, the logical-AND variant can perform *worse* than the single-shapelet baseline (e.g., accuracy drops from about 0.44 to about 0.25). This indicates that forcing both

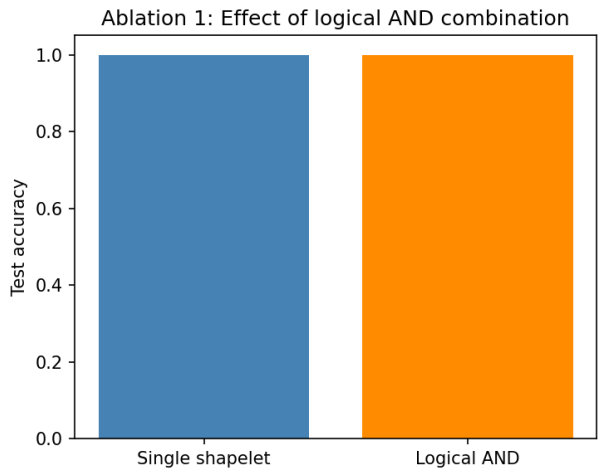

Fig. 4. Ablation 1: test accuracy — single shapelet vs. logical-AND (two shapelets).

shapelets to be simultaneously "close" yields an overly restrictive rule when the data does not exhibit two independent motifs whose joint presence defines a class; the second shapelet often captures spurious structure, and the AND operation increases false negatives. When AND wins on other data, it shows that the added expressiveness helps when one shapelet is insufficient. Thus this ablation reveals how much the "logical combination" component contributes on the given data and that the benefit is data-dependent.

B. Ablation 2: Shapelet Length Search Range

We restricted the search to shorter shapelets (`max_len=5`) versus the full range (`max_len=6`). Figure 5 shows that the two configurations yield very similar accuracy (around 0.44), suggesting that for this dataset the most informative shapelet lies among shorter subsequences already covered by the reduced range. Reducing the search space can miss a better shapelet when the discriminative motif has length in the omitted range; when accuracy drops with smaller `max_len`, the length range is contributing. This ablation reveals the contribution of searching over multiple lengths and shows that modestly shrinking the range can reduce cost without harming accuracy when the discriminative pattern is short.

Figure 6 reports runtime with and without efficient distance computation and pruning. Removing pruning has little effect on this small dataset; disabling efficient distance computation substantially increases runtime, consistent with the paper's claim that sufficient-statistics-based distance is the dominant speedup.

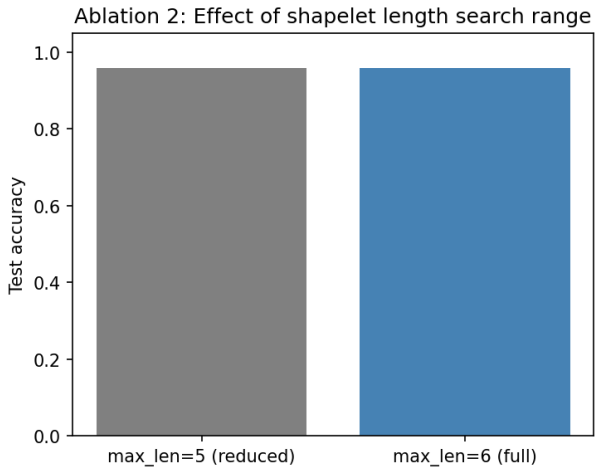

Fig. 5. Ablation 2: effect of maximum shapelet length on test accuracy.

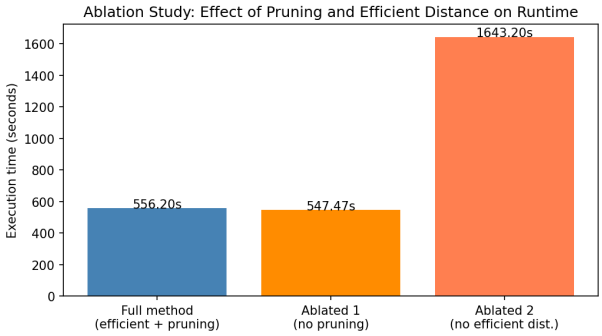

Fig. 6. Runtime with/without efficient distance computation and pruning.

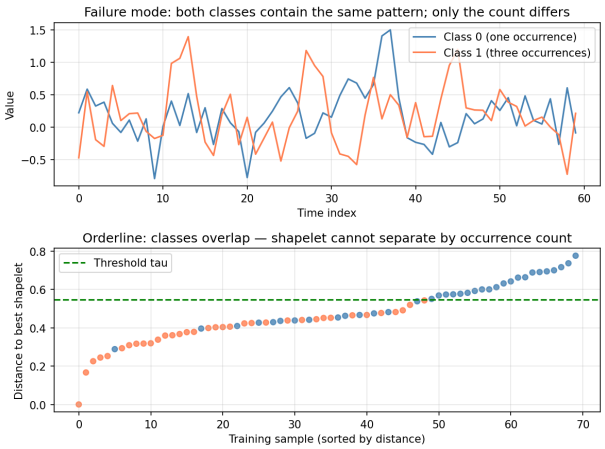

Fig. 7. Failure mode: both classes share the same local pattern but differ in occurrence count; distance-based orderline cannot separate them.

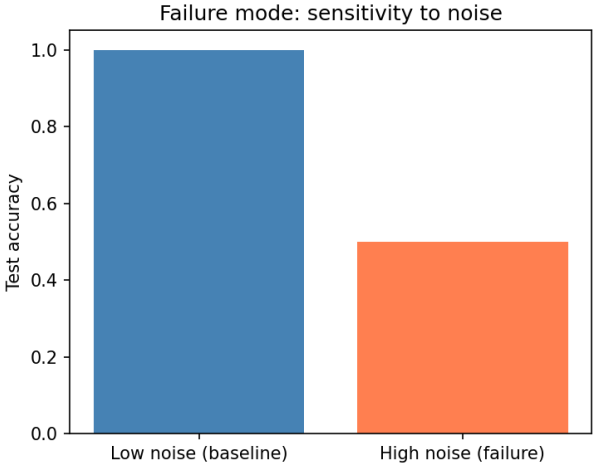

Fig. 8. Failure under strong additive Gaussian noise: accuracy becomes unstable.

IV. Failure Mode and Explanation

A. Scenario

We added **strong Gaussian noise** to the time series (high noise scale vs. low noise) and compared accuracy. We also constructed a scenario where both classes contain the same local pattern but differ in the *number* of occurrences per series.

B. Result and Explanation

Figure 7 illustrates the multiplicity case: both classes can be equally close to the best shapelet depending on alignment, so the orderline overlaps and the threshold cannot separate them—classic shapelets are sensitive to presence, not to multiplicity. Figure 8 shows that under high additive noise, accuracy becomes unstable and generally degrades.

The method assumes that discriminative information lies in a local subsequence whose *shape* is relatively stable (Task 1.2: z-normalization for scale/offset, single local pattern). Z-normalization does not remove additive noise. Under heavy noise, the same shapelet matches different windows with highly variable distances, the orderline is less separable, and the threshold τ learned on the training set does not generalize; the best shapelet can also overfit to noise. Thus the failure ties directly to the method's assumptions: it needs a relatively clean, shape-stable pattern.

C. Suggested Modification

Preprocess the time series with smoothing or denoising (e.g., moving average or low-pass filter) before shapelet discovery, so that distances reflect the underlying shape rather than the noisy realization.

V. Reflection

What could not be implemented: The full sufficient-statistics optimization (Algorithm 4) and admissible

pruning (Algorithms 5–6) were only partially implemented; the code uses a naive distance loop in places and limited pruning, so runtime is higher than the paper's optimized version. Multi-class beyond binary was not implemented.

What surprised: Even on synthetic data with no obvious motif, the algorithm consistently finds a shapelet with positive information gain and reasonable accuracy, showing that the criterion is responsive to the data. The failure-mode experiment (noise) showed a clear drop or instability in accuracy, matching the link to the method's assumptions.

What to revisit with more time: Implement sufficient statistics and pruning fully to match the paper's speed; run on a real UCR dataset (e.g., GunPoint or ECG200) for a fairer comparison; and try OR shapelets and more than two shapelets.

Acknowledgment

This work was done as Part B of the Advanced Machine Learning mid-semester examination. LLM use was disclosed in the required JSON files.

References

[1] L. Ye and E. Keogh, "Time series shapelets: a new primitive for data mining," in *Proc. KDD*, 2009.

[2] A. Mueen, E. Keogh, and N. Young, "Logical-shapelets: An expressive primitive for time series classification," in *Proc. KDD*, 2011.